STAT 20000 Chapter 20
The Chance Error in a Sample Percentage

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- Statistic — the **sample percentage**:
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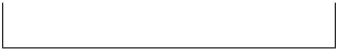
To know how close the sample percentage is from the population percentage, we need a **box model**.

A Box Model



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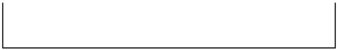

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To sum up,

$\boxed{\text{population} = \text{box}}$ $\boxed{\text{sample} = \text{tickets drawn}}$

Box Model for a SRS

In general, when interested in percentage of subjects in a population with some favored property, a SRS drawn from the population can be analyzed with a box model:

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- A ticket is marked 1 if the subject has the favored property, otherwise 0.
- Number of tickets in the box = population size
- Number of draws = sample size

Box Model for a SRS (Cont'd)

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$$\begin{aligned} &= \text{percentage of } \boxed{1}\text{'s in the sample} \\ &= \frac{\text{count of } \boxed{1}\text{'s in the sample}}{\text{sample size}} \times 100\% \\ &= \frac{\color{red}{\text{sum of draws}}}{\#\text{ of draws}} \times 100\% \end{aligned}$$

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  = percentage of $\boxed{1}$'s in the sample
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  $$= \frac{\text{sum of draws}}{\#\text{ of draws}} \times 100\%$$
- Average of the box = fraction of $\boxed{1}$'s in the box
- SD of the box =

$$\sqrt{(\text{fraction of } \boxed{1} \text{'s in the box}) \times (\text{fraction of } \boxed{0} \text{'s in the box})}$$

The Expected Value for the Sample Percentage

Recall

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The **expected value** for the **sample percentage** is thus

$$\begin{aligned} &\frac{(\#\text{ of draws})(\text{Average of box})}{\#\text{ of draws}} \times 100\% \\ &= \text{Average of box} \times 100\% \\ &= (\text{fraction of } \boxed{1}\text{'s in the box}) \times 100\% \\ &= \boxed{\text{population percentage}} \end{aligned}$$

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When draws are made **with** replacement, we know

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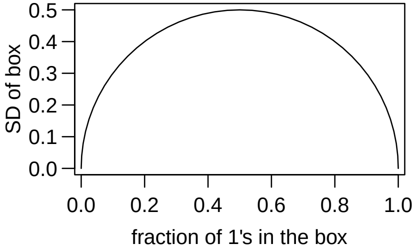
$$\boxed{\text{A larger population does NOT require a larger sample!}}$$

SE for the Sample Percentage (Cont'd)

Recall

$$\text{SD of 0-1 box} = \sqrt{p\,(1-p)},$$

where
$p =$ fraction of $\boxed{1}$'s in the box.



The largest the SD of a 0–1 box could be is ______.

So for a SRS of size 2,500, the largest the SE for the sample percentage could be is ____________________ $= 1\%$.

For a SRS of size a few <u>thousand</u>, the sample percentage is likely to be within a few percent of the population percentage.

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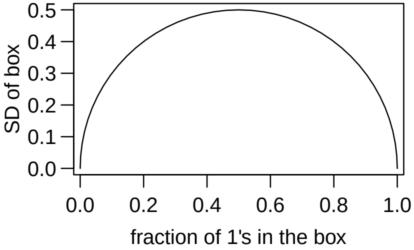

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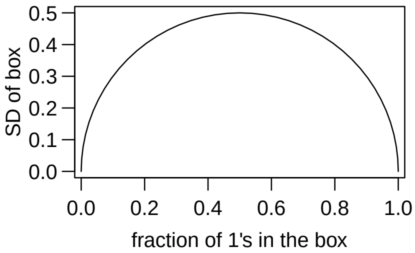

The largest the SD of a 0–1 box could be is <u>0.5</u>.

So for a SRS of size 2,500, the largest the SE for the sample percentage could be is <u>$0.5/\sqrt{2500} = 0.01$</u> $= 1\%$.

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For reasonably large samples, the normal approximation can be used, just as for the sum of the draws.

In particular, the difference between the sample and population percentage is

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Warning: The results (box model, EV, SE, normal approximation) so far are for simple random samples only, NOT applicable to other sampling schemes.

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In a SRS, draws are made **without** replacement. How is our conclusion about the sample average affected?

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- The expected value is __unchanged__.

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- The expected value is $\underline{\text{ unchanged }}$.
- The SE needs to be adjusted

$$\begin{pmatrix}\text{SE when drawing}\\ \textbf{w/o}\text{ replacement}\end{pmatrix} = \begin{pmatrix}\text{correction}\\ \text{factor}\end{pmatrix} \times \begin{pmatrix}\text{SE when drawing}\\ \textbf{w/}\text{ replacement}\end{pmatrix}$$

where the correction factor is

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- The "w/o replacement" SE is smaller since sampling w/o replacement is more efficient. Drawing a ticket a 2nd time tells us nothing new about the box.

Example

You have hired a polling organization to take a SRS from a box of 100,000 tickets, and estimate the percentage of 1's in the box. Unknown to them, the box contains 50% 0's and 50% 1's. How far off should you expect them to be if they draw:

(a) 2,500 (b) 25,000 (c) 100,000 tickets?

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Answer: The SEs for the sample percentage when drawing w/o replacement are

$$\text{(a)}\ \sqrt{\frac{100000-2500}{100000-1}} \times \frac{\sqrt{0.5 \times 0.5}}{\sqrt{2500}} \times 100\% \;=\; 0.987 \times 1\% \approx 1\%$$

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(c) __0__% since all tickets are drawn

Example: Using the Normal Curve

A town has 30000 registered voters, of whom 50% are known to be Democrats. A SRS of 1600 registered voters will be drawn from the population.

Q1: The percentage of Democrats in the sample will be around ________ give or take _________ or so.

Q2: What the chance that the sample percentage is between 49% and 51%?

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- The expected value for the sample percentage = __50%__
- The SE for the sample percentage =

$$\sqrt{\frac{0.5 \times 0.5}{1600}} \times 100\% = \frac{0.5}{40} \times 100\% = 1.25\%$$

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- SU $= \dfrac{51 - 50}{1.25} = 0.8$. Chance = ________

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- correction factor $\approx \sqrt{1 - 1600/30000} = 0.973 \approx 1$
- SU $= \dfrac{51 - 50}{1.25} = 0.8$. Chance = __57.63%__

Summary

For the sample percentage of a SRS drawn from a population,

- expected value = population percentage
- $\text{SE} = \begin{pmatrix}\text{correction}\\ \text{factor}\end{pmatrix} \times \dfrac{\sqrt{p(1-p)}}{\sqrt{\text{sample size}}} \times 100\%$, where $p = (\text{fraction of } \boxed{1}\text{'s in the box})$, and the correction factor is

$$\sqrt{(\text{population size} - \text{sample size})/(\text{population size} - 1)}.$$

- When the sample size is small relative to the population size, the correction factor is nearly 1 and can be ignored.
- The normal approximation can be used to figure chances about a sample percentage; the conversion to standard units is done as

$$z = \frac{\text{sample percentage} - \text{population percentage}}{\text{SE for the sample percentage}}.$$

  The normal approximation can be trusted if the sample size is reasonably large.